

Reduced Conductor Fresnel for Real-Time Rendering

Technical companion to the `fresnel-reduction-validation` repository

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Abstract

This document describes what the repository computes and how each quantity is validated against a full spectral render. The subject is an extension of the non-orthogonal spectral reduction of Fichet, Belcour & Barla (2024), which covers angle-independent operators, to the angle-dependent specular Fresnel of a conductor. The Fresnel operator is factored on the four universal angular basis functions of Belcour, Bati & Barla (2020), giving a 3×3 matrix $\mathbf{M}_F(\theta) = \sum_i b_i(\theta) \Phi_i$ applied to transported radiance. The theory as implemented is in Sections 2 to 4, the validation in Sections 5 and 6, the analysis of the residual error in Section 7, and the limitations in Section 9.

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1 Scope and reading guide

The reduction of Fichet et al. collapses an N -wavelength linear operator into a $K \times K$ matrix ($K = 3$ for XYZ) so that colour transport can run in a tristimulus renderer. Its published form applies to operators that are diagonal in wavelength and independent of direction: albedo, transmittance, fluorescent reradiation. The specular reflectance of a metal is neither separable that way nor negligible: the Fresnel term $F(\theta, \lambda)$ depends jointly on angle and wavelength, and it is the only wavelength-dependent factor of a microfacet BRDF.

The idea validated here is to make it separable. Belcour et al. (2020) showed that for any physical conductor the angular curve $\cos \theta \mapsto F(\cos \theta)$ lies in a four-dimensional subspace with universal basis functions $b_0..b_3$. Applying that per wavelength writes $F(\theta, \lambda) = \sum_i c_i(\lambda) b_i(\theta)$, and each $c_i(\lambda)$ is exactly the kind of operator Fichet’s reduction handles. Four fixed 3×3 matrices per material, four scalar weights per pixel.

The repository answers three questions with numbers:

1. Is the *angular* decomposition accurate for gold, inside a renderer? (Yes: $\Delta E_{2000} \approx 0.02$ – 0.04 against a spectral reference.)
2. Is the *reduced matrix* accurate? (Not to target: a systematic tint of $\Delta E_{2000} \approx 2$ – 2.7 under smooth illuminants, 0.8 under a fluorescent lamp.)
3. Where does that tint come from, and what would remove it? (It is the metameric black of the illuminant folded into the visible by the material; anchoring the reconstruction operator on smooth spectra removes it at no runtime cost, and the matrix form is what keeps that stable over bounces.)

Throughout, spectra are column vectors on a 1 nm grid from 380 to 780 nm ($N = 401$), matrices act on the left, and θ is the angle between the incident direction and the microfacet normal (half vector).

2 Spectral reduction of a wavelength-only operator

2.1 Basis, dual basis, reduced operator

Let $\mathbf{S} \in \mathbb{R}^{N \times 3}$ hold the CIE 1931 colour-matching functions as columns. A spectrum $\mathbf{E}$ has coordinates $\mathbf{c} = \mathbf{S}^T \mathbf{E}$; the dual basis

$$\tilde{\mathbf{S}} = \mathbf{S} (\mathbf{S}^T \mathbf{S})^{-1}, \quad \mathbf{S}^T \tilde{\mathbf{S}} = \mathbf{I}_3, \quad (1)$$

reconstructs from coordinates the unique spectrum in $\text{span } \tilde{\mathbf{S}}$ with those coordinates. For a wavelength-diagonal operator $\text{diag}(f)$ the reduced operator is

$$\Phi = \mathbf{S}^\top \text{diag}(f) \tilde{\mathbf{S}} \in \mathbb{R}^{3 \times 3}, \quad \mathbf{c}_{\text{out}} = \Phi \mathbf{c}_{\text{in}}, \quad (2)$$

exact whenever $\mathbf{E} \in \text{span } \tilde{\mathbf{S}}$ and otherwise in error by the part of $\mathbf{E}$ the basis cannot see. Section 7 makes that part precise; it accounts for the entire residual.

2.2 Normalisation, and why it differs from the fluorescence work

`compute_cmf.py` samples $\mathbf{S}$ from Mitsuba’s own internal CMF table (`mi.cie1931_xyz`) and divides all three columns by the *single* sum of the $\bar{y}$ column. This differs from the per-column normalisation used in the sibling reflectance/fluorescence repository. Per-column normalisation applies three different gains and would introduce a fixed per-channel tint relative to the spectral film’s own XYZ conversion, which uses one common factor. `check_conventions.py` proves the stored $\mathbf{S}$ equals Mitsuba’s table exactly and that the hardcoded IEC 61966-2-1 XYZ→sRGB matrix equals `mi.xyz_to_srgb` to 8.6×10^{-5} . Both checks exit non-zero on failure. The two repositories’ $\mathbf{S}$ matrices are therefore not interchangeable, and Φ is invariant to a common rescaling of $\mathbf{S}$ but not to a per-column one.

3 Conductor Fresnel and its angular decomposition

3.1 Exact Fresnel

A conductor is described by a complex index $\hat{n}(\lambda) = \eta(\lambda) + i\kappa(\lambda)$. For unpolarised light at incidence θ ,

$$a^2 + b^2 = \sqrt{(\eta^2 - \kappa^2 - \sin^2 \theta)^2 + 4\eta^2 \kappa^2}, \quad (3)$$

$$R_s = \frac{a^2 + b^2 - 2a \cos \theta + \cos^2 \theta}{a^2 + b^2 + 2a \cos \theta + \cos^2 \theta}, \quad R_p = R_s \frac{a^2 + b^2 - 2a \sin \theta \tan \theta + \sin^2 \theta \tan^2 \theta}{a^2 + b^2 + 2a \sin \theta \tan \theta + \sin^2 \theta \tan^2 \theta}, \quad (4)$$

$$F(\theta, \lambda) = \frac{1}{2} (R_s + R_p). \quad (5)$$

`fresnel_tools.py` implements exactly this (it is the evaluation routine from the authors’ helper module, reduced to the functions used). `load_ior.py` resamples gold’s η, κ from a two-block CSV in micrometres onto the 1 nm grid and asserts plausible values at 550 nm.

3.2 The four universal basis functions

Belcour et al. (2020) build a corpus of normalised Fresnel curves $\tilde{F}(\cos \theta) = (F(\cos \theta) - F(1))/(1 - F(1))$ over a large set of dielectric and conductor IORs, and extract by a two-pass SVD a basis $\{b_0 \equiv 1, b_1, b_2, b_3\}$ on a 1000-point $\cos \theta$ grid such that every physical curve is well approximated by four coefficients (Figure 1). **This repository does not re-derive the basis:** `extract_basis.py` parses the table `fresnel_curves[1000][4]` from the authors’ published Mitsuba plugin source (`data/belcour2020/fresnel_curves.hpp`) together with the two probe indices their coefficient solve uses. b_1 is the SVD-extracted first mode, not the Schlick $(1 - \cos \theta)^5$ it resembles; the two differ by up to 0.12 at grazing, and using Schlick in one place and the extracted b_1 in another was a bug found and fixed during the work.

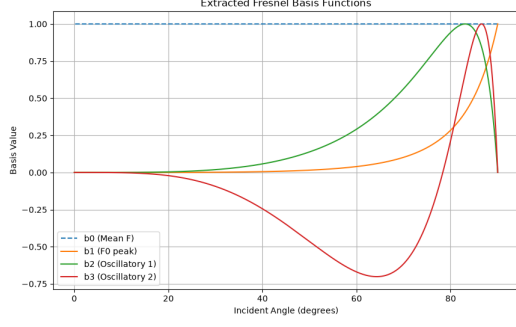


Figure 1: The four universal angular bases read from the authors’ table. $b_0 \equiv 1$; b_3 is negative over most of its range, which matters for how it is stored on the GPU (Section 8).

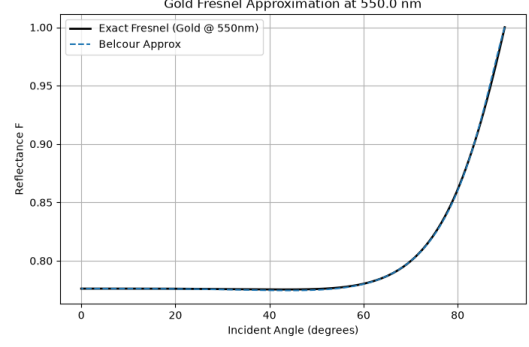


Figure 2: Exact conductor Fresnel of gold at 550 nm and its four-coefficient reconstruction.

3.3 Per-wavelength coefficients

At a fixed λ the decomposition is

$$F(\cos \theta, \lambda) \approx \sum_{i=0}^3 c_i(\lambda) b_i(\cos \theta), \quad c_0 = F(1), \quad c_1 = \max(1 - c_0, 0), \quad (6)$$

and c_2, c_3 are the solution of the 2×2 system obtained by matching the residual $F - c_0 - c_1 b_1$ at two probe angles ($\cos \theta_1 = 0.1201$, $\cos \theta_2 = 0.4344$, the arg-max positions of b_2 and b_3), solved by Cramer’s rule exactly as in the authors’ `fresnelToCoeffs()`. `compute_coeffs.py` does this for each of the 401 wavelengths and checks the reconstruction at 550 nm against the exact curve over all 1000 angles: **maximum absolute error** 0.0036 (Figure 2), guarded by an assertion at 0.01.

4 The synthesis: reduced Fresnel matrices

4.1 Separating angle from wavelength

Each coefficient function $c_i(\lambda)$ is a wavelength-only operator, so Eq. (2) applies to it:

$$\Phi_i = \mathbf{S}^T \text{diag}(c_i(\lambda)) \tilde{\mathbf{S}} \in \mathbb{R}^{3 \times 3}, \quad i = 0..3, \quad (7)$$

computed in `compute_phi.py` as `(S.T * c[i]) @ Sb` without materialising the diagonal. The reduced Fresnel operator at any angle is then

$$\boxed{\mathbf{M}_F(\theta) = \sum_{i=0}^3 b_i(\theta) \Phi_i} \quad \mathbf{c}_{\text{out}} = \mathbf{M}_F(\theta) \mathbf{c}_{\text{in}}. \quad (8)$$

Angle enters only through four scalars and wavelength only through four constant matrices. The operator is therefore cheap (one basis fetch, four 3×3 multiply-adds, one matrix-vector product) and can be preintegrated. The b_i are universal; only the Φ_i change with the material. `compute_phi.py` checks that under a flat white illuminant at $\cos \theta = 0.5$ the reflected luminance is positive, below the incident luminance, and larger than the reflected Z (gold is yellow-orange).

4.2 Split-sum preintegration for image-based lighting

For a rough surface under distant lighting the specular integral factors, in the split-sum approximation of Karis (2013), into a prefiltered environment term and a preintegrated BRDF term.

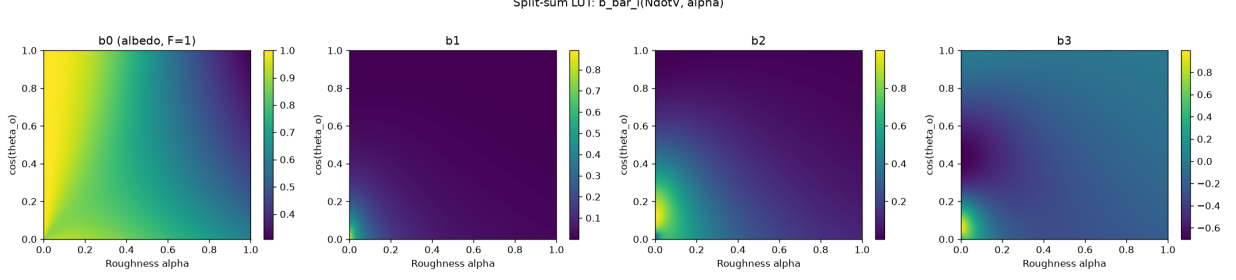


Figure 3: The preintegrated weights $\bar{b}_i(\cos \theta_o, \alpha)$. Channel 0 is the achromatic GGX albedo; channels 1–3 concentrate at grazing view and low roughness, where the Fresnel curve departs from its normal-incidence value.

Because the b_i are scalar and universal, the BRDF term is preintegrated once per basis:

$$\bar{b}_i(\cos \theta_o, \alpha) = \int_{\Omega} b_i(\omega_i \cdot \mathbf{h}) \frac{D(\mathbf{h}; \alpha) G_2(\omega_i, \omega_o; \alpha)}{4(\omega_o \cdot \mathbf{n})} d\omega_i, \quad \bar{\mathbf{M}}_F(\cos \theta_o, \alpha) = \sum_i \bar{b}_i \Phi_i. \quad (9)$$

`compute_split_sum.py` estimates each $\bar{b}_i$ on a 256×256 grid in $(\cos \theta_o, \alpha)$ with 4096 Halton samples per cell, importance-sampling the GGX visible normal distribution (Heitz 2018). The per-sample weight reduces to $G_1(\omega_i)(\omega_i \cdot \mathbf{h})/(\omega_o \cdot \mathbf{h})$ because the VNDF density cancels D and $G_1(\omega_o)$ algebraically. The b_0 channel is the ordinary $F=1$ GGX directional albedo and serves as a sanity check (asserted near 1 at normal incidence and low roughness, and below 1 elsewhere). Figure 3 shows the four channels. No render in this repository exercises the LUT: every comparison uses the exact-angle basis of Eq. (8) at $\alpha = 0.05$, where the mirror and the split-sum forms coincide. This matches the real-time target chosen for the first engine scene, a mirror-like conductor lit by an environment map sampled once along the reflected direction, which needs only Eq. (8). The split-sum form is the extension to rough surfaces; it was deferred because the engine does not expose its prefiltered environment, and its intrinsic limits are listed in Section 9.

4.3 Working in linear sRGB

Φ_i acts in XYZ, while a tristimulus renderer (Mitsuba’s RGB variants, Godot) transports linear sRGB. Rather than converting radiance twice per pixel, the colour change is folded once into each matrix,

$$\Psi_i = \mathbf{M}_{\text{rgb} \leftarrow \text{xyz}} \Phi_i \mathbf{M}_{\text{xyz} \leftarrow \text{rgb}}, \quad (10)$$

so that $\mathbf{M}_F$ assembled from Ψ_i applies directly to sRGB radiance. `audit_tint.py` verifies the two routes agree to 5.6×10^{-16} for a single bounce (pure linearity), and `export_godot.py` verifies the fold inverts back to Φ_i to 3.3×10^{-16} .

5 Validation renderers

All renders share one scene: a unit gold sphere, GGX with $\alpha = 0.05$, lit by an environment, seen by a narrow 16° camera, 1024×1024 pixels, 2048 samples per pixel, Mitsuba’s `direct` integrator (one bounce with next-event estimation and multiple importance sampling). Three renderers produce three images per illuminant.

Reference (`render_reference.py`). Mitsuba’s `llvm_ad_spectral` variant with the stock `roughconductor` and gold’s η, κ given as spectral files. Each ray packet carries four wavelengths; the film projects onto XYZ with the CMF table of Section 2.2. It serves as ground truth.

Spectral path (`render_ours.py`). The same spectral variant, with a custom Python BSDF that keeps everything of the stock microfacet model except the Fresnel term, which it replaces per wavelength by the four-coefficient reconstruction of Section 3.3, solving the 2×2 system on the fly for the wavelengths of each packet. No reduction is involved: this isolates the angular decomposition alone.

Reduced-matrix path (`render_ours_direct.py`). The `llvm_ad_rgb` variant with a custom integrator that re-implements `direct` (emitter sampling, BSDF sampling, power-heuristic weights) but replaces the scalar Fresnel multiply by the matrix product $L_{\text{out}} = \mathbf{M}_F(\cos \theta_d) L_{\text{in}}$ on the actual transported radiance, using the pre-folded Ψ_i . It is the real-time formulation.

The $F=1$ trick. Neither custom path re-implements GGX. Each instantiates a stock `roughconductor` with a dummy index ($\eta = 0$, $\kappa = 10^5$, hence $F \approx 1$ at all angles) and uses it as an achromatic geometry term and VNDF sampler, then multiplies by its own Fresnel. The angle fed to the Fresnel is the half-vector angle $\cos \theta_d = \boldsymbol{\omega}_i \cdot \mathbf{h}$, recomputed as `dot(si.wi, normalize(si.wi + wo))`. Basis values are gathered from the 1000-entry table at `round(999 cos  $\theta_d$ )` in both paths.

5.1 The reference confound and the four illuminants

The Uffizi environment map is an RGB image. In a spectral variant, Mitsuba must invent a spectrum for every RGB pixel and does so with Jakob & Hanika (2019) upsampling; the reduced path implicitly uses the dual-basis metamer $\tilde{\mathbf{S}}\mathbf{c}$ instead. Two arbitrary metamers of the same RGB pixel: the residual between reference and reduced render mixes the reduction error with a disagreement that has nothing to do with the material. The Uffizi comparison shows this directly: its error map is non-zero on the background, where no surface is involved (Figure 4).

`make_illuminants.py` therefore produces three deconfounded lighting conditions, each rendered by all three paths:

- **Uffizi, achromatic:** the luminance of the HDRI replicated to $R=G=B$. Keeps the angular structure; an achromatic pixel maps to a near-flat spectrum under both upsamplings.
- **D65:** a constant emitter with the CIE D65 spectral power distribution as a file. Smooth. Zero upsampling ambiguity.
- **FL1:** the same with CIE FL1, a spiky fluorescent lamp.

The two constant-emitter scenes have no environment structure: the sphere is lit uniformly from every direction, so the renders show a nearly flat disc against a flat background, as in a furnace test. This removes geometry from the comparison and leaves only the material’s spectral response, at the cost of the scene. The SPD files are normalised so that the spectral film sees luminance $Y = 1$ for a perfect reflector, which requires $\int E \bar{y} d\lambda = \int \bar{y} d\lambda$, not $= 1$.

5.2 Metric

`compare.py` reports per-pixel RMSE on linear HDR values and per-pixel CIEDE2000 on Reinhard-tonemapped sRGB converted to CIELAB. The ΔE_{2000} is display-referred; it is the number quoted everywhere below. NaN pixels (present in the spectral reference as isolated black speckles) are zeroed in all three images before comparison.

6 Results

The spectral path is at the noise floor under every illuminant; its error maps are black except for the reference’s NaN speckles. The angular decomposition is therefore not the limiting factor, and

Illuminant	Nature	spectral path		reduced matrix	
		ΔE_{2000}	RMSE	ΔE_{2000}	RMSE
Uffizi HDRI	RGB, upsampled (confounded)	0.020	0.0008	2.107	0.030
Uffizi, achromatic	RGB, deconfounded	0.020	0.0008	2.036	0.029
D65	spectral, smooth	0.044	0.0009	2.700	0.032
FL1	spectral, spiky	0.041	0.0008	0.783	0.011

Table 1: Mean CIEDE2000 (display-referred) and mean linear-HDR RMSE against the spectral reference, gold sphere, $\alpha = 0.05$, 2048 spp. Target for the reduced path was a mean ΔE_{2000} below 1. All values regenerated from this repository; they match the original runs to three decimals.

the error of the reduced path comes from the reduction itself.

The reduced path has a systematic, multiplicative tint. Under D65 the sphere is a uniform disc and the error map is a uniform ≈ 2.7 over it. Under the structured Uffizi lighting the error follows the radiance: highest on the bright arch reflection, lowest in the dark regions. Visually the reduced gold is less saturated and greener than the reference (Figures 4–5).

The dependence on the illuminant runs against intuition: the spiky fluorescent FL1 is handled three times better than the smooth daylight D65. Section 7 explains why.

7 Diagnosis of the residual tint

`audit_tint.py` reproduces the tint offline in pure linear algebra, without renderer, Monte Carlo or tonemapping, and writes its output to `results/audit_tint.txt`. The tint appears there, so it is a property of the mathematics rather than of the renderer.

7.1 The tint is the mathematics, not the renderer

At normal incidence under D65, the exact colour $\mathbf{S}^\top(F \circ \mathbf{E})$ and the reduced colour $\Phi(\mathbf{S}^\top \mathbf{E})$ convert to linear sRGB as

$$\text{exact} = [1.025, 0.722, 0.340], \quad \text{reduced} = [0.961, 0.766, 0.329],$$

while the rendered pixels at the sphere centre are $[1.0226, 0.7199, 0.3412]$ and $[0.9563, 0.7626, 0.3272]$. Three-decimal agreement with a closed-form computation: about -6% red, $+6\%$ green.

7.2 Metameric black

Write the illuminant as the part the basis sees plus the part it cannot:

$$\mathbf{E} = \tilde{\mathbf{S}} \mathbf{S}^\top \mathbf{E} + \mathbf{B}_E, \quad \mathbf{B}_E = (\mathbf{I} - \tilde{\mathbf{S}} \mathbf{S}^\top) \mathbf{E}, \quad \mathbf{S}^\top \mathbf{B}_E = 0. \quad (11)$$

$\mathbf{B}_E$ is the *metameric black* of $\mathbf{E}$ (Wyszecki): invisible to the observer, and therefore invisible to the reduced pipeline. The single-bounce error is exactly

$$\mathbf{S}^\top(F \circ \mathbf{E}) - \Phi(\mathbf{S}^\top \mathbf{E}) = \mathbf{S}^\top(F \circ \mathbf{B}_E), \quad (12)$$

verified numerically to 4×10^{-16} . A flat material leaves $\mathbf{B}_E$ invisible; gold’s wavelength-varying F modulates it back into the visible. For D65, $\|\mathbf{B}_E\|/\|\mathbf{E}\| = 0.55$: over half of the illuminant’s energy, in L_2 , is black to the dual basis. A spiky spectrum such as FL1 happens to have less of its energy in that complement, which is why it is handled better.

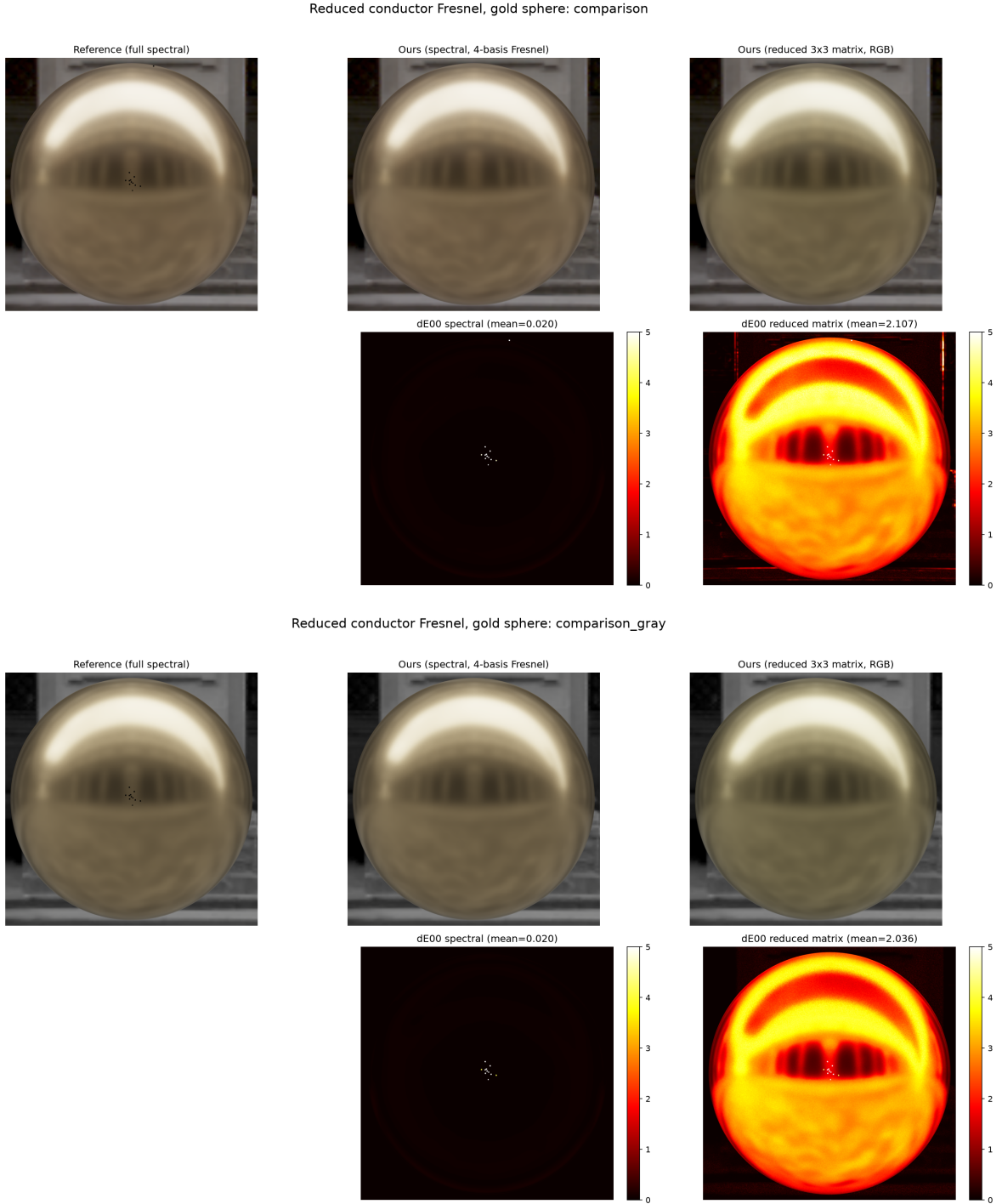


Figure 4: Uffizi HDRI (top) and its achromatic version (bottom). Left to right: reference, spectral path, reduced-matrix path; below, ΔE_{2000} maps. Note the non-zero background error in the top row, absent in the bottom row.

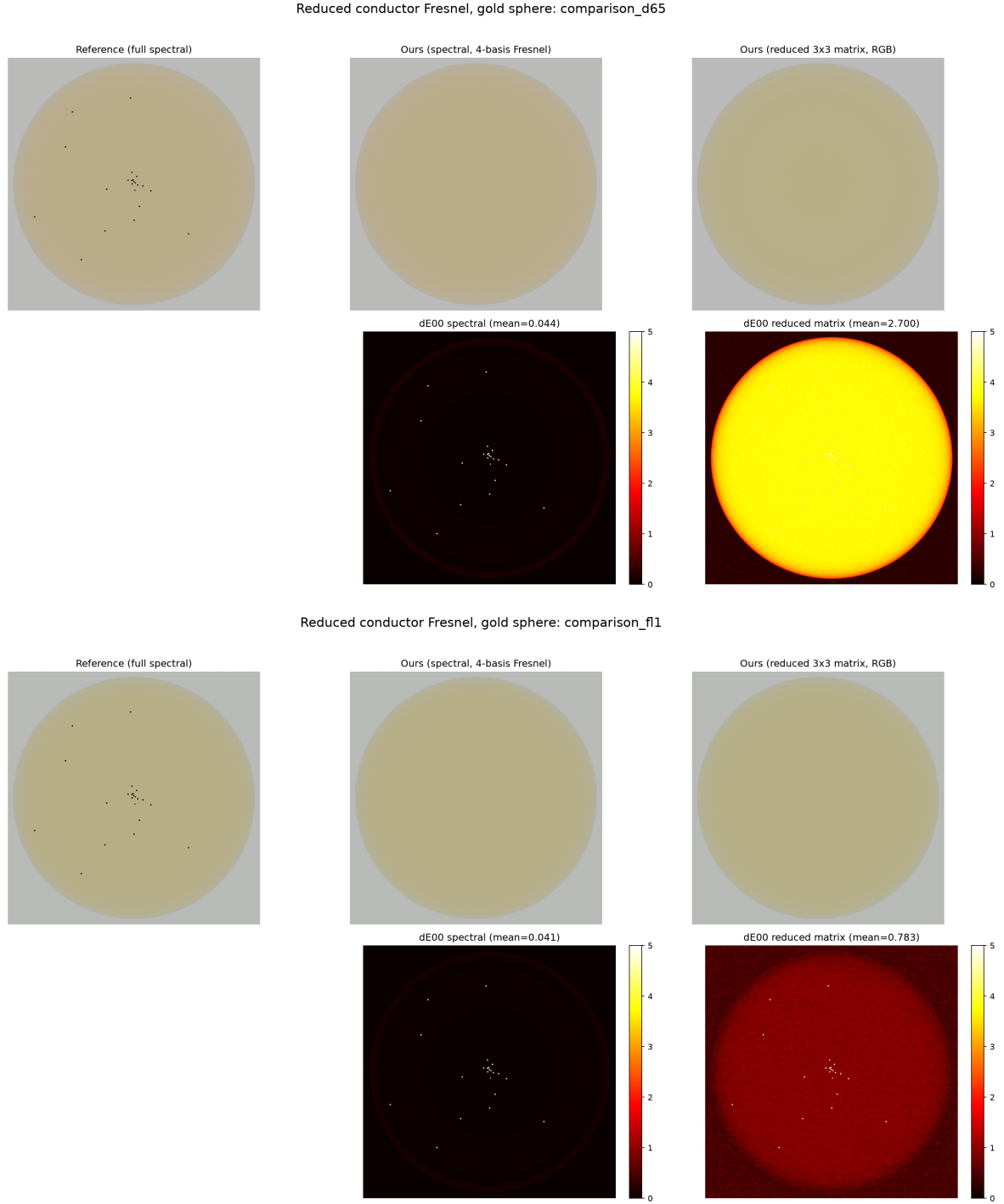


Figure 5: Natively spectral constant emitters: D65 (top) and FL1 (bottom). Under D65 the reduced path is uniformly wrong by ≈ 2.7 ; under FL1 by ≈ 0.8 .

7.3 The reconstruction operator is a free design axis

Nothing in Eq. (2) forces the reconstruction to be $\tilde{\mathbf{S}}$. Any $\mathbf{R} \in \mathbb{R}^{N \times 3}$ with $\mathbf{S}^\top \mathbf{R} = \mathbf{I}_3$ gives a valid reduced operator $\Phi = \mathbf{S}^\top \text{diag}(f) \mathbf{R}$, exact on $\text{span } \mathbf{R}$. The dual basis is the choice with minimal norm; it is not the choice with minimal error on the spectra one actually meets. `audit_tint.py` compares several, at five angles, against the exact spectral colour (Table 2). These ΔE_{2000} values are computed on un-tonemapped $O(1)$ colours and are *not* numerically comparable to Table 1; only their ranking and the closed-form reproduction above carry over.

Reconstruction $\mathbf{R}$ / baseline	E	A	D50	D65	D75	FL1	FL2	FL4
[A] dual basis $\tilde{\mathbf{S}}$ (this work)	5.52	2.45	5.93	7.00	7.57	1.69	0.43	0.71
[A4] same, 4-basis $\mathbf{M}_F$	5.49	–	–	6.97	–	1.66	–	–
[C] daylight $S_0 S_1 S_2$ anchor	0.58	0.33	0.00	0.00	0.00	4.80	5.42	3.61
[D] anchored on D65, FL1, A	3.81	0.00	5.04	0.00	4.69	0.00	0.85	3.23
[I] Mallett–Yuksel sRGB primaries	0.00	0.37	0.54	0.99	1.27	3.83	5.14	3.64
[E] diagonal, flat-weighted	0.00	4.44	2.47	1.24	3.67	5.29	2.36	2.83
[G] Belcour 2020, RGB-averaged IOR	1.51	4.72	3.86	1.55	2.61	3.87	1.33	2.63

Table 2: Mean ΔE_{2000} over five angles, single bounce, gold, against the exact spectral colour. Rows [A]–[I] are matrix operators differing only in $\mathbf{R}$; [E] and [G] are per-channel (diagonal) baselines of the kind engines use. Exact zeros are exact by construction.

The [A] and [A4] rows agree to 0.03, so the angular decomposition adds nothing to the error, in line with Section 6. An $\mathbf{R}$ anchored on the CIE daylight components makes the reduction exact on every daylight illuminant and near-exact on the flat one, at the same runtime cost, since only the Φ_i change. It is worse on fluorescent lamps, where the dual basis does better. No single 3-dimensional $\mathbf{R}$ is good everywhere; the choice fixes the family of illuminants on which the reduction is exact.

7.4 Why a matrix, and not a diagonal

The [E] and [G] rows already show that a per-channel Fresnel is wrong by 1–5 ΔE_{2000} once the illuminant is not the one it was averaged under. The decisive comparison is over depth. Ward’s prefiltering (row [F] of the audit) sets $\rho_k = (\mathbf{S}^\top (F \circ \mathbf{E}))_k / (\mathbf{S}^\top \mathbf{E})_k$ and is therefore *exact at one bounce, by construction*. Table 3 iterates each operator three times at $\cos \theta = 0.707$.

Operator	Illuminant	1 bounce	2 bounces	3 bounces	cond $\mathbf{R}$
[A] dual basis	D65	8.28	9.25	8.10	1.0
[C] daylight anchor	D65	0.00	1.14	1.92	8.4
[I] Mallett–Yuksel	D65	1.12	0.32	1.47	–
[F] diagonal, Ward-prefiltered	D65	0.00	4.77	9.83	–
[E] diagonal, flat	D65	1.37	2.70	6.81	–
[D] anchored on three lamps	D65	0.00	7.32	14.72	49.7
[H] PCA of ten lamps	D65	0.59	15.91	49.29	113
[A] dual basis	FL1	2.12	2.15	1.47	1.0
[C] daylight anchor	FL1	5.83	8.52	8.93	8.4
[F] diagonal, Ward-prefiltered	FL1	0.00	5.91	11.92	–

Table 3: ΔE_{2000} after n applications of the operator, against n exact spectral reflections. The diagonal prefilter is exact once and degrades linearly; the smooth-anchored matrix stays below 2; naively anchored reconstructions are ill-conditioned and diverge.

The matrix formulation is what allows an anchored reconstruction to survive transport depth: the off-diagonal terms carry the inter-channel coupling that a diagonal operator fabricates at the

second bounce. It is also a warning: an anchor chosen by fitting a handful of lamps ([D], [H]) has a badly conditioned $\mathbf{R}$ and explodes, so the smoothness of the anchor family matters more than its fit at one bounce.

8 Export to the engine

`export_godot.py` writes `results/godot_gold.json` with the four pre-folded Ψ_i (row-major, applied as $\text{out}_r = \sum_c \Psi_{rc} L_c$) and the four basis arrays b_i on the uniform 1000-point $\cos \theta$ grid. The export also fixes the following:

- The basis is shipped as the *full sampled arrays*, packed into a 1000×1 RGBA32F texture and fetched at $u = \cos \theta$, rather than as a polynomial fit. $b_3 \in [-0.70, 1.0]$ is negative over most of its range, so an 8-bit texture would silently clamp it.
- A Godot Basis bound to a `mat3` uniform reads columns; the row-major JSON must be uploaded column by column.

Before writing, the script checks that the fold of Eq. (10) inverts back to Φ_i (3.3×10^{-16}) and that the four-basis assembly $\sum_i b_i \Phi_i$ agrees with the exact reduced operator $\mathbf{S}^T \text{diag}(F(\cos \theta)) \tilde{\mathbf{S}}$ recomputed from the IOR at five angles (maximum entry difference 2.2×10^{-3} , the four-basis fit error). It exits non-zero if either fails. The runtime cost is one RGBA fetch, four 3×3 multiply-adds and one matrix-vector product per pixel.

9 Limitations and caveats

1. **The reduced path misses its target.** Mean ΔE_{2000} of 2.0–2.7 under smooth illuminants against a target below 1. The cause is understood (Section 7) and a fix is identified (a smooth-anchored $\mathbf{R}$) but *not implemented in the render pipeline*: all rendered results use the dual basis. Regenerating Φ_i with the daylight anchor and re-rendering is the immediate next step.
2. **The split-sum LUT is unvalidated in a render.** It inherits every limit of the split-sum approximation: bilinear blur at low α and grazing view, per-textel Monte Carlo variance, isotropic roughness only, the separable $G_2 \approx G_1 G_1$ error frozen into the table, single scattering (GGX energy loss at high roughness), and the distant-lighting assumption that integrates ω_i out.
3. **The reference has NaN pixels.** Isolated black speckles in the spectral reference (absent in the spectral path, which shares its geometry code) are zeroed before comparison and inflate the maximum ΔE_{2000} , not the mean. Their origin in Mitsuba’s spectral `roughconductor` at low roughness was not investigated.
4. **Two ΔE_{2000} conventions.** `compare.py` measures tonemapped, display-referred colour over the whole image including the background; `audit_tint.py` measures un-tonemapped colour at five angles on the material alone. Magnitudes do not transfer between Tables 1 and 2.
5. **The Uffizi comparison is confounded** by the upsampling disagreement of Section 5.1; its background error is the proof. Quote the three deconfounded rows.
6. **Single bounce in the renderer.** `render_ours_direct.py` applies the matrix once and clamps negatives in sRGB per bounce; a path-traced extension should carry the throughput as an ordered product of matrices in XYZ and clamp only at the sensor. Table 3 is offline algebra, not a render.

7. **The basis is not re-derived.** It is the authors’ published table, read verbatim; its training corpus (99% dielectrics, 1% conductors) bounds its fidelity on exotic metals.
8. **Normalisation differs from the fluorescence repository** (Section 2.2). Do not reuse one repository’s $\mathbf{S}$ or $\tilde{\mathbf{S}}$ in the other.
9. **Prior art.** Reduced-coordinate transport with matrix-valued throughput is already in Fichet et al. (2024, §4.1) and, earlier, in Percy (1993). What is new here is only the angular factorisation $\mathbf{M}_F(\theta) = \sum_i b_i(\theta)\Phi_i$ and its matrix-valued split-sum.

10 Reproduction

```
make venv          # python3 -m venv .venv ; pip install -r requirements.txt
make precompute    # load_ior, extract_basis, compute_coeffs, compute_cmf, compute_phi
make check         # check_conventions + export_godot (both exit 1 on failure)
make audit         # audit_tint -> results/audit_tint.txt
make illuminants   # scene/uffizi_gray.exr, scene/illum_{d65,fl1}.spd
make render        # 12 renders: {reference, ours, ours_direct} x {uffizi, gray, d65, fl1}
make compare       # 4 comparison figures with their Delta-E numbers
make splitsum      # results/split_sum.npy and its figure (Monte Carlo, all cores)
make doc           # this document (latexmk), after ‘make figures’
```

All scripts resolve `data/`, `scene/` and `results/` relative to their own location (`scripts/_paths.py`) and run from any directory. Renders are deterministic for a fixed Mitsuba version and seed. Pinned versions: Python 3.12, Mitsuba 3.9.0, Dr.Jit 1.4.0, numpy 2.5.0, scipy 1.18.0, matplotlib 3.11.0, colour-science 0.4.7, imageio 2.37.3, scikit-image 0.26.0, pandas 3.0.3, joblib 1.5.3. The `.exr` renders and the achromatic environment map are not committed; every PNG and every small artefact in `results/` is.

A File map

B Symbols

References

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- E. Heitz. *Sampling the GGX Distribution of Visible Normals*. JCGT, 2018.
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Path	Role
scripts/load_ior.py	Section 3; gold η, κ on the 1 nm grid
scripts/extract_basis.py	Section 3.2; $b_0..b_3$ and the probe angles from the authors' table
scripts/compute_coeffs.py	Section 3.3; $c_i(\lambda)$, fit check
scripts/compute_cmf.py	Section 2.2; $\mathbf{S}, \tilde{\mathbf{S}}$
scripts/compute_phi.py	Eq. (7); $\Phi_0.. \Phi_3$
scripts/compute_split_sum.py	Section 4.2; $\bar{b}_i$ LUT
scripts/check_conventions.py	Section 2.2; CMF and matrix agreement with Mitsuba
scripts/export_godot.py	Section 8; Ψ_i , basis, checks
scripts/audit_tint.py	Section 7; every number in Tables 2–3
scripts/make_illuminants.py	Section 5.1; deconfounded lighting assets
scripts/render_reference.py	Section 5; spectral ground truth
scripts/render_ours.py	Section 5; spectral path, four-basis Fresnel
scripts/render_ours_direct.py	Section 5; reduced-matrix path
scripts/compare.py	Section 6; metrics and figures
scripts/fresnel_tools.py	exact conductor Fresnel (authors' helper, reduced)
data/gold_refractive_index.csv	complex IOR of gold
data/belcour2020/fresnel_curves.hpp	the universal basis table (authors' Mitsuba plugin)
scene/	Mitsuba scenes, spectral IOR and illuminant files, Uffizi HDRI

Symbol	Shape	Meaning
$\mathbf{S}, \tilde{\mathbf{S}}$	$N \times 3$	CMF basis (common $\bar{y}$ normalisation) and its dual
$\mathbf{R}$	$N \times 3$	any reconstruction with $\mathbf{S}^\top \mathbf{R} = \mathbf{I}$
$\eta(\lambda), \kappa(\lambda)$	N	real and imaginary index of gold
$F(\theta, \lambda)$	scalar	exact conductor Fresnel
$b_i(\cos \theta)$	1000 samples	universal angular bases, $i = 0..3$
$c_i(\lambda)$	N	per-wavelength coefficient of b_i
Φ_i, Ψ_i	3×3	reduced coefficient operator in XYZ, and pre-folded in linear sRGB
$\mathbf{M}_F(\theta)$	3×3	assembled reduced Fresnel, Eq. (8)
$\bar{b}_i(\cos \theta_o, \alpha)$	256×256	preintegrated weights, Eq. (9)
$\mathbf{B}_E$	N	metameric black of illuminant $\mathbf{E}$